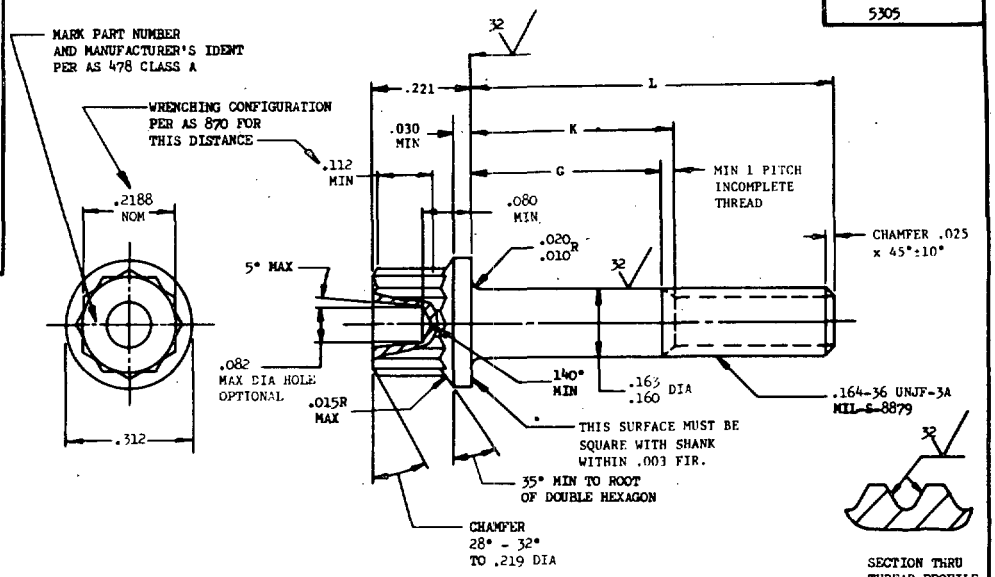
MS9931

FED. SUP CLASS
5305



SECTION THRU
THREAD PROFILE

PART NO.	L	G MIN	K MAX	APPROX WEIGHT LB/100
MS9931-10	.750	.052	.172	.672
MS9931-11	.812	.114	.234	.708
MS9931-12	.875	.177	.297	.746
MS9931-13	.938	.240	.360	.783
MS9931-14	1.000	.302	.422	.821
MS9931-15	1.062	.364	.484	.858
MS9931-16	1.125	.427	.547	.895
MS9931-17	1.188	.490	.610	.933
MS9931-18	1.250	.542	.672	.970
MS9931-19	1.312	.614	.734	1.008
MS9331-20	1.375	.677	.797	1.045
MS9331-21	1.438	.740	.860	1.082
MS9931-22	1.500	.802	.922	1.120

PART NO.	L	G MIN	K MAX	APPROX WEIGHT LB/100
MS9931-23	1.562	.864	.984	1.157
MS9931-24	1.625	.927	1.047	1.195
MS9931-25	1.688	.990	1.110	1.232
MS9931-26	1.750	1.052	1.172	1.269
MS9931-27	1.812	1.114	1.234	1.307
MS9931-28	1.875	1.177	1.297	1.344
MS9931-29	1.938	1.240	1.360	1.382
MS9931-30	2.000	1.302	1.422	1.419

1. SHANK SHALL BE STRAIGHT WITHIN .003 TOTAL PER INCH OF SCREW LENGTH.
2. THE CONCENTRICITY OF THREAD PD IN RELATION TO THE SHANK SHALL BE WITHIN .006 FIR.
3. THE CONCENTRICITY OF THE SHANK IN RELATION TO THE WASHER FACE DIAMETER AND DOUBLE HEXAGON OD SHALL BE WITHIN .005 FIR.
4. MATERIAL: STEEL AMS 6322.
5. HARDNESS: ROCKWELL C26-32.
6. FINISH: CADMIUM PLATE AMS 2400. DIMENSIONS SPECIFIED ARE AFTER PLATING.
7. MANUFACTURING SPECIFICATION: AMS 7452 EXCEPT HEAD SHALL BE UPSET.
8. HEAD TO SHANK FILLET SHALL BE COLD WORKED.
9. MAGNETIC PARTICLE INSPECTION PER AMS 2640 AFTER PLATING.
10. SURFACE TEXTURE: USAS B46.1-1962. UNLESS OTHERWISE SPECIFIED, SURFACES TO BE 125 MICROINCHES EXCEPT UPSET HEAD.
11. BREAK SHARP EDGES .003-.015 UNLESS OTHERWISE SPECIFIED.
12. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: LINEAR DIMENSIONS ±.010, ANGULAR DIMENSIONS ±5°.
13. DO NOT USE UNASSIGNED PART NUMBERS.

AS & AMS ARE SOCIETY OF AUTOMOTIVE ENGINEERS, INC. PUBLICATIONS.
THIS STANDARD WAS DEVELOPED COOPERATIVELY WITH THE MILITARY SERVICES BY THE SAE AEROSPACE PART STANDARDS DIVISION.

APPROVED 30 Dec 70 REVISED

User activities: Navy - MC, OS

Review activities: Air Force - 82
Navy - AS

This military standard is approved by the Department of the Air Force and the Bureau of Naval Weapons and is mandatory for use by these activities. All other military activities are required to employ this standard where suitable.

P.A.	TITLE	MILITARY STANDARD
AIR FORCE - 11 Other Cust NAVY - AS	SCREW, MACHINE - DOUBLE HEXAGON EXTENDED WASHER HEAD, FULL SHANK, STEEL AMS 6322, CADMIUM PLATED, .164-36 UNJF-3A	MS9931
PROCUREMENT SPECIFICATION AMS-7452	SUPERSEDES:	SHEET 1 OF 1

DD FORM 1 DEC 67 672-1 (Limited coordination)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

☆ U. S. GOVERNMENT PRINTING OFFICE: 1971-433-696/4806

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:28Z
Check the source to verify that this is the current version before use.